Understanding the Performance of AI Algorithms in Text-Based Emotion Detection for Conversational Agents

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***Abstract*—This paper provides an overview of how emerging conversational AI systems are being utilized for various business applications such as customer support, educating users with technology-based educational tools, and assisting consumers with day-to-day tasks through digital assistants. A majority of these conversational AI's do not incorporate any type of emotional context into their user input processing, resulting in their lack of empathetic response and robotic nature. Our research will demonstrate an emotion-aware conversational system that utilizes a BERT based emotion recognition model [1] and will be incorporated with a context-aware, Groq API enabled large language model. The user's text will be analyzed based on a transformer-based emotion classification model [2] o detect the user's emotional state (happiness, sadness, anger, fear, surprise neutrality). The detected user's emotional state will be utilized during the generation of the chatbot's response. An additional conversation memory module will preserve all of the dialogue history for each pair (user and AI assistant) to assist with maintaining the context of the entire conversation during multiple interactions. Lastly, a React-based web-based interface will be developed to show detected user's emotions in real-time on top of the chatbot's response. Experimental evaluations demonstrate that providing emotion detection in addition to a conversational AI can increase conversational coherence, increase the user's perception of the AI's ability to demonstrate empathy, and increase the level of user engagement, compared to standard chatbots.**

***Index Terms*—Emotion Detection, Conversational AI, BERT, Large Language Models, Natural Language Processing, Emotion-Aware Chatbots, Groq API**

I. Introduction

Emotional aspects of communication are intrinsic to humans. The tone or feel/ambience of conversations will affect how they play out, as will how a person feels toward what was said or intended [3]. Traditional chatbot systems mostly rely on semantic analysis but leave out the human emotional signaling, resulting in responses that may make sense in context, but will not create an emotionally supportive atmosphere in some areas (such as mental health support, customer service,tutoring) [4].

Recent technological developments in natural language processing allow machines to quantify emotional impressions through improved accuracy over time.For example, BERT not only created a two-way contextual representation of the words in a sentence together with their meanings, but it also enabled accurate identification of a text's emotion based on this representation [1]. At the same time, LLM have demonstrated their ability to generate meaningful text that appears as though it has been generated through an emotionally supportive and coherent manner in an online conversational context [5].

The research described in this paper proposes an Emotion-Aware AI with Conversational capabilities, which are capable of detecting the emotional state of the user messaging based on the information returned by a BERT-based classifier. This information will provide the foundation for preparing the appropriate way to write a response to the user's message, based on this specific user message's emotional state as well as the entire history of the conversation, permitting the generation of both semantically, as well as emotionally, appropriate response messages from the LLM, when responding to user messages.

The contributions of this research include:

- Six emotional categories (happy, sad, angry, fearful, surprised or neutral) are categorized by a BERT-based emotion detection component.
- Detected emotions are integrated into an LLM prompt to generate a contextually appropriate response with respect

to emotional states.

- Multi-turn dialogue context is preserved by the conversation memory component.
- The graphical web interface is built using React and Tailwind CSS, and includes real-time emotion visualization.

II. Related Work

Initially, emotion recognition typically used a lexicon approach to perform sentiment analysis with the use of an emotion lexicon. This was effective when trying to classify simple/textbook cases, but it did not work for expressions that relied on context or were too complex. The next wave of emotion recognition included the use of machine learning classifiers like SVM and Naive Bayes. These classifiers primarily relied on hand-crafted features to build their emotion representations, such as bag-of-words and TF-IDF, which posed a limitation to the emotion recognition system since these features were created manually.

The emergence of deep learning architectures such as LSTM and CNNs allowed for improved performance when trying to capture word relationships based upon sequence [6], or by extracting local textual features that were relevant to emotion [7]. The emergence of transformer-based architectures like the BERT model [1] was a notable advancement for emotion detection, as its use of bidirectional context has allowed for a far better understanding of context across an entire sentence at once. Since then, BERT-based systems have consistently achieved state-of-the-art results when performing emotion classification [8].

Recent advances in technology have coincided with an increasing amount of research on emotion-sensitive dialogue systems. In 2017, Zhou et al. [11] proposed an emotional chatting machine that produced responses based on emotive labels. In the same year, Rashkin et al. [12] created a dataset for developing empathetic dialogue and created models capable of creating empathetically appropriate responses to inputs. These systems often do not provide a long-term memory of the conversation, and they don't tightly couple the emotions they detect into their large language model (LLM) response generation pipelines. Our proposed system overcomes both of those limitations.

III. System Architecture

The proposed system has three layers of architecture: a front end user interface; a back-end processing engine; an artificial intelligence (AI) intelligence layer.

Frontend Interface: he front end of the proposed system is developed in React using Tailwind CSS. The front end displays messages exchanged with the chatbot in a conversation panel, with a visual indicator showing the emotion associated with each chatbot response. A sidebar allows for the creation of chat session histories, and users have the ability to create new chat sessions or delete the existing session histories from the sidebar. User messages sent to the backend will use a HTTP Rest API to send messages to the backend.

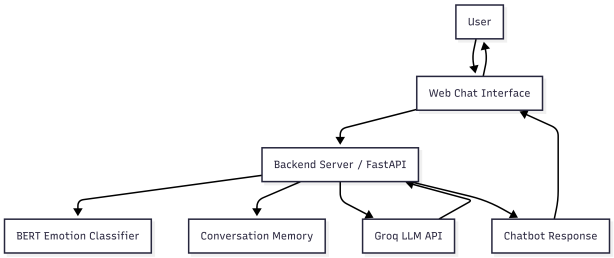

Fig. 1. Overall System Architecture of the Emotion-Aware Conversational AI System

Backend Processing Engine: The back end of the proposed system is created using FastAPI and manages all the system components. When the back end receives a user message, it sends the message to the Bert emotion classifier to identify the emotion associated with the user's message; the back end then retrieves conversation history for all messages associated with that chat session; constructs an emotion aware prompt; and sends the emotion-aware prompt to the Groq API to get a response from a large language model of the user's preferred choice; to then save the returned response in memory, then return the response to the front end for display in the UI.

AI Intelligence Layer: This layer includes the BERT emotion classifier, a Large Language Model (LLM) that generates responses on Groq's infrastructure, and a conversation memory manager that maintains the dialogue state throughout the conversation.

IV. Methodology

A. Text Preprocessing

User inputs are first passed through tokenization, followed by normalisation and noise removal. Performing these transformations on the user input helps ensure BERT's predictions of emotions will be accurate and the user inputs will provide consistent and clean text from BERT for inference.

B. Emotion Detection using BERT

The Emotion Classifier utilizes a pre-trained BERT model that has been further trained on an Emotion Labelled Dataset to classify the user input as one of six emotions: happy, sad, angry, fear, surprise or neutral [9]. During the classification of the full length of the user input, BERT's multi-head self-attention performs self-attention on the entire token sequence using both forward and backward directions resulting in creating a context-based representation of the user input. BERT creates a context-based classification head that maps the context-based BERT Representations into a class of probability scores, over each emotion class available.

$$Emotion = \arg\max_{i} \left(P(e_1), P(e_2), \ldots, P(e_n)\right)$$

where $P(e_i)$ is the probability of the input belonging to emotion class e_i. For example, the input *"I feel really stressed about my exams"* may yield a Sad classification with a confidence of 0.87.

C. Context-Aware Response Generation

On the backend, after the emotion classification has been completed, a prompt is constructed using the current user message, along with any conversation history, as well as the emotion classification of the current user message, and sent to the Groq API. The emotion classification has been used to give the LLM direction on how to create a response that is tone appropriate for the current user message. For example, if the user message has been classified as sad, empathetic and supportive responses would be generated as opposed to neutral and informational responses [12].

D. Conversation Memory Management

Each dialogue turn (user message, chatbot response, detected emotion, and timestamp) is saved in a memory module, therefore, when the chatbot generates a new response it includes the entire prior conversation as context for the LLM prompt. This allows the chatbot to refer back to previous exchanges, continue the same topic from earlier exchanges, and not give contradictory responses to users for multi-turn interactions.

V. Implementation

The system is based on the full stack architecture with three clear layers being frontend, backend, and AI model. The technology stack used is provided in Table I.

TABLE I
Implementation Technology Stack

Component	Technology
Frontend Framework	React
UI Styling	Tailwind CSS
Backend Framework	FastAPI (Python)
Emotion Detection	BERT Transformer
Language Model	Groq API LLM
Conversation Memory	Session-based Storage

The entire end to end workflow is illustrated in Figure 2. A user submits a message through the web-based user interface (UI) and this message is routed through REST API to the backend. The backend processes the incoming text and sends it to the BERT classifier for analysis. The resulting emotion label and conversation history retrieved from the database are combined with the user-submitted message creating a prompt for the Groq API to generate the response, which is stored in memory and sent back to the UI a) as the chatbot's response, and b) with the emotion label for presentation.

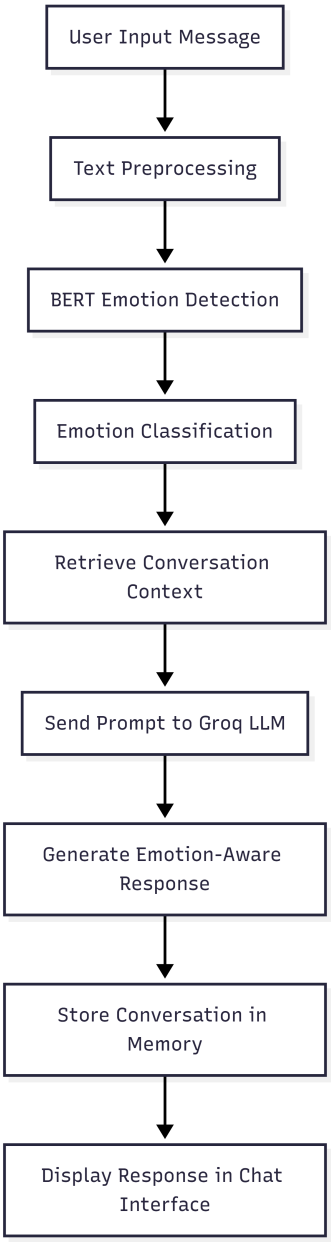

Fig. 2. Step-by-Step Workflow of the Emotion-Aware Chatbot System

VI. Experimental Results

A. Emotion Classification Performance

The BERT classifier was evaluated using precision, recall, and F1-score on an established emotion dataset with all six emotion categories represented. Table II presents the results of the evaluation of the BERT emotion classification model..

The best results for determining emotions via two distinct language groups can be attributed to clearly defined emotional states such as Happy, Sad, and Neutral. The results based on those three states existed within their own unique linguistic patterns. The results for the ambiguous categories of Fear and Surprise were slightly less effective, but still acceptable. The results of the study correlates with previous BERT-based research on emotional classification [10].

B. Conversational Response Quality

The qualities of each response were assessed based on emotional appropriacy and how coherent the conversation was between each of the test subgroups. In the cases evaluated, the chatbot was able to determine the appropriate tone to use given the determined emotional state detected. Responses to Sad

TABLE II
EMOTION CLASSIFICATION PERFORMANCE SUMMARY

Emotion	Precision	Recall	F1-Score
Happy	High	High	High
Sad	High	High	High
Angry	High	Moderate	High
Fear	Moderate	High	Moderate
Surprise	Moderate	Moderate	Moderate
Neutral	High	High	High

emotion messages gave empathetic responses from the chatbot; Happy emotion messages resulted in positive/encouraging responses from the chatbot. The disabling of the memory module resulted in loss of coherence in conversation between multiple turns, this is a clear indication of the importance of preserving context for high-quality multi-turn conversations.

VII. WEB APPLICATION INTERFACE

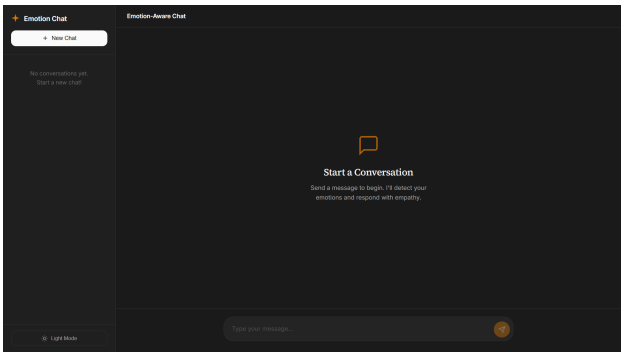

Fig. 3. Emotion-Aware Chatbot Web Application Interface

The web interface of the chatbot application is shown on Figure 3, the application interface provides user and chatbot messages in a conversation panel format for users convenience. Each response generated by the chatbot generates an emotion indicator in real-time indicating the detected emotion *Detected Emotion: Sad* combined with an appropriate emoji, allows the users to assess the emotional interpretation the chatbot generated as a result of their initial message. In the far left sidebar are links to previous sessions of conversation for easy navigation to each session. There are also links to create a new chat or delete any previous conversation session at any time. In providing the user with a real-time emotion label displayed for each response, users have complete visibility into the actions or output of the tools used by the chatbot system.

Users can navigate through prior conversations using the left hand navigation bar. Users may initiate new conversations or delete prior conversations at any point. The live emotion labels help to make clear what the system is doing and create a more engaging and informative user experience.

VIII. CONCLUSION

This research presents a conversational AI that considers and interacts with its users' emotional states using a BERT based emotion detection system [1] nd a Groq API based LLM context aware response generation. By detecting the emotional states of user messages and then conditioning the LLM on both the detected emotional state and the history of the conversation, this chatbot produces empathetic and contextually appropriate responses that traditional chatbots cannot [11].

The system was developed as a full stack application, which consists of a React based front-end, FastAPI based back-end, and session based memory. The accuracy of emotional classifications and the comparative quality of responses and conversational coherence were improved when using both the emotional detection and memory modules.

In the future, the researchers plan to investigate how to implement multimodality in emotional detection by using voice and facial expression analysis, develop personalized emotional models for individual users, and utilize reinforcement learning to optimize the adaptive response mechanism [12].

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